

Highlights

,

-
-
-

★

$a,*,1$, $b,2$

a
, , , ,
 b
, , , ,

ARTICLE INFO	ABSTRACT
<i>Keywords:</i>	Here goes the abstract

1.

*
*Corresponding author
ORCID(s):
1

Figure 1:

Table 1

2.

CRediT authorship contribution statement

: : : .

References

- [1] J. L. Araus, J. E. Cairns, Field high-throughput phenotyping: the new crop breeding frontier, *Trends Plant Sci.* 19 (2014) 52–61.
- [2] J. L. Araus, S. C. Kefauver, M. Zaman-Allah, M. S. Olsen, J. E. Cairns, Translating high-throughput phenotyping into genetic gain, *Trends Plant Sci.* 23 (2018) 451–466.
- [3] C. Zhao, et al., Crop phenomics: current status and perspectives, *Front. Plant Sci.* 10 (2019) 714.
- [4] L. Li, Q. Zhang, D. Huang, A review of imaging techniques for plant phenotyping, *Sensors* 14 (2014) 20078–20111.
- [5] G. J. Rebetzke, J. Jimenez-Berni, R. A. Fischer, D. M. Deery, D. J. Smith, High-throughput phenotyping to enhance the use of crop genetic resources, *Plant Sci.* 282 (2019) 40–48.
- [6] R.-F. Wang, H.-R. Qu, W.-H. Su, From sensors to insights: technological trends in image-based high-throughput plant phenotyping, *Smart Agric. Technol.* 12 (2025) 101257.
- [7] S. Paulus, Measuring crops in 3D: using geometry for plant phenotyping, *Plant Methods* 15 (2019) 103.
- [8] A. Paturkar, G. S. Gupta, D. Bailey, Making use of 3D models for plant physiognomic analysis: a review, *Remote Sens.* 13 (2021) 2232.
- [9] M. S. Akhtar, Z. Zafar, R. Nawaz, M. M. Fraz, Unlocking plant secrets: a systematic review of 3D imaging in plant phenotyping techniques, *Comput. Electron. Agric.* 222 (2024) 109033.
- [10] N. Harandi, B. Vandenbergh, J. Vankerschaver, S. Depuydt, A. Van Messem, How to make sense of 3D representations for plant phenotyping: a compendium of processing and analysis techniques, *Plant Methods* 19 (2023) 60.
- [11] J. Li, et al., A survey on 3D reconstruction techniques in plant phenotyping: from classical methods to neural radiance fields (NeRF), 3D Gaussian splatting (3DGS), and beyond, *Plant Phenomics* 26 (2025) 100137.
- [12] J. Qi, et al., Multiscale phenotyping of grain crops based on three-dimensional models: a comprehensive review of trait detection, *Comput. Electron. Agric.* 237 (2025) 110597.
- [13] T. T. Nguyen, D. C. Slaughter, N. Max, J. N. Maloof, N. Sinha, Structured light-based 3D reconstruction system for plants, *Sensors* 15 (2015) 18587–18612.
- [14] S. Debnath, M. Paul, T. Debnath, Applications of LiDAR in agriculture and future research directions, *J. Imaging* 9 (2023) 57.
- [15] M. Vazquez-Arellano, D. Reiser, D. S. Paraforos, M. Garrido-Izard, H. W. Griepentrog, 3-D reconstruction of maize plants using a time-of-flight camera, *Comput. Electron. Agric.* 145 (2018) 235–247.
- [16] J. L. Schonberger, J.-M. Frahm, Structure-from-motion revisited, in: *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2016, pp. 4104–4113.
- [17] Y. Furukawa, J. Ponce, Accurate, dense, and robust multiview stereopsis, *IEEE Trans. Pattern Anal. Mach. Intell.* 32 (2010) 1362–1376.
- [18] W. Sheng, et al., MVS-Pheno: a portable and low-cost phenotyping platform for maize shoots using multiview stereo 3D reconstruction, *Plant Phenomics* 5 (2023) 0143.
- [19] Y. Wang, S. He, H. Ren, W. Yang, R. Zhai, 3DPhenoMVS: a low-cost 3D tomato phenotyping pipeline using 3D reconstruction point cloud based on multiview images, *Agronomy* 12 (2022) 1865.
- [20] B. Mildenhall, et al., NeRF: representing scenes as neural radiance fields for view synthesis, *Commun. ACM* 65 (2022) 99–106.
- [21] T. Muller, A. Evans, C. Schied, A. Keller, Instant neural graphics primitives with a multiresolution hash encoding, *ACM Trans. Graph.* 41 (2022) 102.
- [22] A. Chen, Z. Xu, A. Geiger, J. Yu, H. Su, TensorRF: tensorial radiance fields, in: *ECCV*, 2022, pp. 333–350.
- [23] M. Tancik, et al., Nerfstudio: a modular framework for neural radiance field development, in: *ACM SIGGRAPH Conf. Proc.*, 2023, pp. 1–12.

- [24] J. Afful, et al., Evaluating Neural Radiance Fields for 3D Plant Geometry Reconstruction in Field Conditions, *Plant Phenomics* 6 (2024) Article 0235.
- [25] A. Jignasu, et al., Plant geometry reconstruction from field data using neural radiance fields, *AAAI Workshop on AI for Agriculture and Food Systems*, 2023.
- [26] H.-B. Choi, J.-K. Park, S. H. Park, T. S. Lee, NeRF-based 3D reconstruction pipeline for acquisition and analysis of tomato crop morphology, *Front. Plant Sci.* 15 (2024) 1439086.
- [27] B. Kerbl, G. Kopanas, T. Leimkuhler, G. Drettakis, 3D Gaussian Splatting for real-time radiance field rendering, *ACM Trans. Graph.* 42 (2023) 139.
- [28] B. Huang, et al., 2D Gaussian Splatting for geometrically accurate radiance fields, *ACM Trans. Graph.* 43 (2024) 1–17.
- [29] A. Guedon, V. Lepetit, SuGaR: Surface-Aligned Gaussian Splatting for efficient 3D mesh reconstruction and high-quality mesh rendering, in: *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2024, pp. 5354–5363.
- [30] J. Ma, et al., Plant3R: Fusing 3D feature learning with Gaussian splatting to enhance wheat plant 3D reconstruction precision, *Plant Phenomics* (2026). doi:10.1016/j.plaphe.2026.100200.
- [31] P. Shen, X. Jing, W. Deng, H. Jia, T. Wu, PlantGaussian: exploring 3D Gaussian splatting for cross-time, cross-scene, and realistic 3D plant visualization and beyond, *Crop J.* 13 (2025) 607–618.
- [32] L. Jiang, J. Sun, P. W. Chee, C. Li, L. Fu, Cotton3DGaussians: multiview 3D Gaussian splatting for boll mapping and plant architecture analysis, *Comput. Electron. Agric.* 234 (2025) 110293.
- [33] J. Li, K. Zhu, Q. Zhang, D. Chen, Q. Sun, Z. Li, Object-centric 3D Gaussian splatting for strawberry plant reconstruction and phenotyping, *Smart Agric. Technol.* 13 (2026) 101810.
- [34] J.-H. Lin, T.-T. Lin, From 3DGS scenes to plant traits: a scalable extraction and segmentation framework for muskmelon phenotyping, *Front. Plant Sci.* 17 (2026) 1783465.
- [35] W. Song, et al., IPENS: Interactive unsupervised framework for rapid plant phenotyping extraction via NeRF-SAM2 fusion, *Plant Phenomics* (2025). doi:10.1016/j.plaphe.2025.100106.
- [36] M. Ye, M. Danelljan, F. Yu, L. Ke, Gaussian grouping: segment and edit anything in 3D scenes, in: *ECCV*, 2024, pp. 162–179.
- [37] J. Cen, et al., Segment any 3D Gaussians, in: *Proc. AAAI Conf. Artif. Intell.*, volume 39, 2025, pp. 1971–1979.
- [38] A. Kirillov, et al., Segment Anything, in: *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2023, pp. 3992–4003.
- [39] N. Ravi, et al., SAM 2: Segment Anything in images and videos, 2024. arXiv:2408.00714.
- [40] T. Cheng, et al., YOLO-World: real-time open-vocabulary object detection, in: *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2024, pp. 16901–16911.
- [41] W. Shi, R. van de Zedde, H. Jiang, G. Kootstra, Plant-part segmentation using deep learning and multi-view vision, *Biosyst. Eng.* 187 (2019) 81–95.
- [42] D. Li, et al., PlantNet: a dual-function point cloud segmentation network for multiple plant species, *ISPRS J. Photogramm. Remote Sens.* 184 (2022) 243–263.
- [43] R. Du, Z. Ma, P. Xie, Y. He, H. Cen, PST: plant segmentation transformer for 3D point clouds of rapeseed plants at the podding stage, *ISPRS J. Photogramm. Remote Sens.* 195 (2023) 380–392.
- [44] X. Wu, et al., Point Transformer V3: simpler faster stronger, in: *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2024, pp. 4840–4851.
- [45] B. Curless, M. A. Levoy, A volumetric method for building complex models from range images, in: *SIGGRAPH*, 1996, pp. 303–312.
- [46] W. E. Lorensen, H. E. Cline, Marching Cubes: a high resolution 3D surface construction algorithm, in: *SIGGRAPH*, 1987, pp. 163–169.
- [47] A. Hore, D. Ziou, Image quality metrics: PSNR vs. SSIM, in: *Int. Conf. Pattern Recognit.*, 2010, pp. 2366–2369.
- [48] Z. Wang, A. C. Bovik, H. R. Sheikh, E. P. Simoncelli, Image quality assessment: from error visibility to structural similarity, *IEEE Trans. Image Process.* 13 (2004) 600–612.
- [49] R. Zhang, P. Isola, A. A. Efros, E. Shechtman, O. Wang, The unreasonable effectiveness of deep features as a perceptual metric, in: *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2018, pp. 586–595.
- [50] S. Pertuz, D. Puig, M. A. García, Analysis of focus measure operators for shape-from-focus, *Pattern Recognit.* 46 (2013) 1415–1432.
- [51] Z. Fan, et al., LightGaussian: unbounded 3D Gaussian compression with 15x reduction and 200+ FPS, *Adv. Neural Inf. Process. Syst.*, 2024.
- [52] J. C. Lee, et al., Compact 3D Gaussian representation for radiance field, *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2024.
- [53] W. Liu, et al., EfficientGS: streamlining Gaussian Splatting for large-scale high-resolution scene representation, 2024. arXiv:2404.12778.
- [54] G. Fang, B. Wang, Mini-splatting: representing scenes with a constrained number of Gaussians, in: *ECCV*, 2024, pp. 165–181.